3. Linear Programming

3.1 Formulate and Solving

The given decision variables are TXV, EVA, CCS, COM, CON, REC that have an impact on the objective function. The objective function is given using profit contribution.

Two Constrains are - Available times (in minutes) for Shaper, Grinder, Binder, Cables and Electronics. Committed Demand for EVA of 500 Nos.

	TXV	EVA	CCS	COM	CON	REC				
Decision Variable	1,369.89	500.00	-	148.52	-	-				
Objective Function	182.00	116.00	206.00	193.00	124.00	175.00	335,984.86		Minutes Available	
Shaper	23.00	31.00	16.00	44.00	33.00	26.00	53,542.47	<=	66,660.00	
Grinder	32.00	39.00	24.00	45.00	32.00	20.00	70,020.00	<=	70,020.00	
Binder	31.00	47.00	60.00	18.00	56.00	57.00	68,640.00	<=	68,640.00	
Cables.and.Electronics	-	22.00	46.00	27.00	-	-	15,010.11	<=	72,000.00	

Table 6

3.1.1 Answer Report

Objective Cell (Max)

Cell	Name	Original Value	Final Value
$O$5	Objective Function	335,984.86	335,984.86

Variable Cells

Cell	Name	Original Value	Final Value	Integer
$I$4	Decision Variable TXV	1,369.89	1,369.89	Contin
$J$4	Decision Variable EVA	500.00	500.00	Contin
$K$4	Decision Variable CCS	-	-	Contin
$L$4	Decision Variable COM	148.52	148.52	Contin
$M$4	Decision Variable CON	-	-	Contin
$N$4	Decision Variable REC	-	-	Contin

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
$O$6	Shaper	53,542.47	$O$6<=$Q$6	Not Binding	13,117.53
$O$7	Grinder	70,020.00	$O$7<=$Q$7	Binding	-
$O$8	Binder	68,640.00	$O$8<=$Q$8	Binding	-
$O$9	Cables.and.Electronics	15,010.11	$O$9<=$Q$9	Not Binding	56,989.89
$J$4	Decision Variable EVA	500.00	$J$4>=$B$27	Binding	-

Table 7

3.2 Objective Function and its Interpretation

3.2.1 Sensitivity Report

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
$I$4	Decision Variable TXV	1369.89	0.00	182	150.39	7.54
$J$4	Decision Variable EVA	500.00	-128.48	116	128.48	1E+30
$K$4	Decision Variable CCS	0.00	-20.87	206	20.87	1E+30
$L$4	Decision Variable COM	148.52	0.00	193	14.54	87.32
$M$4	Decision Variable CON	0.00	-119.48	124	119.48	1E+30
$N$4	Decision Variable REC	0.00	-31.27	175	31.27	1E+30

Table 8

3.2.2 Interpretation

- From the data sheet we can prove that “Objective Coefficient” ranges would be “Profit Contribution” with respect to given components.
- From the Sensitivity report, it is understood that the components with high profit contribution is targeted with additional constraint of making Evaporator to a limit 500 which is the committed demand.

  Example: Evaporator demand is firstly met to an extent of 500, next component targeted is Thermostatic Expansion Valve which shares second highest profit contribution and subsequently next is Compressor which shares the third highest profit contribution.

3.3 Right-Hand-Side Ranges and its Interpretation

3.3.1 Sensitivity Report

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$O$6	Shaper	53542.47	0	66660	1E+30	13117.53
$O$7	Grinder	70020.00	3.305250305	70020	11308.69	3923.87
$O$8	Binder	68640.00	2.459096459	68640	3801.25	24932.00
$O$9	Cables.and.Electronics	15010.11	0	72000	1E+30	56989.89

Table 9

3.3.2 Interpretation

- From the data sheet we can prove that “Right Hand Side” ranges would be “Total Resource Availability (in minutes)” with respect to given processes.
- From the Sensitivity report, it is understood that there are still some unutilised resources available, which can be optimised for the better utilisation of working capital and resources.

  Example: Grinder and Binder are utilised with no Slacks generated, but for Shaper, Cable & Electronics, management can reduce the operational hours and manpower employed for Shaper, Cables & Electronics and they can make a strategy for operation of Grinders and Binder for better utilisation of resources.

3.4 Addition of 1hr to Grinder Time

	TXV	EVA	CCS	COM	CON	REC			
Decision Variable	1,369.89	500.00	-	148.52	-	-			
Objective Function	182.00	116.00	206.00	193.00	124.00	175.00	335,984.86		Minutes Available
Shaper	23.00	31.00	16.00	44.00	33.00	26.00	53,542.47	<=	66,660.00
Grinder	32.00	39.00	24.00	45.00	32.00	20.00	70,020.00	<=	70,020.00
Binder	31.00	47.00	60.00	18.00	56.00	57.00	68,640.00	<=	68,640.00
Cables.and.Electronics	-	22.00	46.00	27.00	-	-	15,010.11	<=	72,000.00

Table 10. Before Adding 1hr to Grinding Time

	TXV	EVA	CCS	COM	CON	REC			
Decision Variable	1,368.57	500.00	-	150.79	-	-			
Objective Function	182.00	116.00	206.00	193.00	124.00	175.00	336,183.17		Minutes Available
Shaper	23.00	31.00	16.00	44.00	33.00	26.00	53,612.06	<=	66,660.00
Grinder	32.00	39.00	24.00	45.00	32.00	20.00	70,080.00	<=	70,080.00
Binder	31.00	47.00	60.00	18.00	56.00	57.00	68,640.00	<=	68,640.00
Cables.and.Electronics	-	22.00	46.00	27.00	-	-	15,071.43	<=	72,000.00

Table 11. After Adding 1hr to Grinding Time

The addition of 1hr to grinder time will helps the organisation to get additional profit and will adjust the production of Thermostatic Expansion Valve and Compressor, shadow price will affect the profit that is addition of 1hr to grinder will add profit to the organisation. It also helps with the utilisation of Shaper.